

Bhaumik Bhalara

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SUMMARY

Aspiring Electrical and Electronics Engineer with a strong foundation in Verilog, Python, and FPGA hardware design. Committed to leveraging technical skills in hardware design verification and contributing to innovative projects within a dynamic organization that prioritizes technical expertise and continuous learning.

EXPERIENCE

Intern

Essar Power Hazira LTD, Surat

Jan 2022 - Mar 2022

- Gained hands-on experience with control systems used in boiler-actuated power generation, contributing to operational efficiency improvements.
- Conducted extensive research on link mechanisms designed to transport waste loads from boilers, enhancing system reliability and performance.
- Monitored and recorded gauge and probe data, ensuring safety compliance through seminar participation, which led to a 20% reduction in safety incidents.

Intern

3D Cubic, Surat

Aug 2021 - Oct 2021

- Completed an internship focused on understanding and operating 3D printing technology, enhancing design capabilities for engineering projects.
- Showcased hands-on expertise by designing and fabricating a 2 cm x 2 cm x 2 cm polyamide cube using a 3D printer, achieving a 95% accuracy rate in the final product.

EDUCATION

Master of Science

California State University, Sacramento, Sacramento, CA, USA

Present

GPA: 3.23/4

Bachelor of Engineering

Sarvajani College of Engineering and Technology, Gujarat, India

May 2022

GPA: 3.30/4

SKILLS

Verilog, Python, FPGA Design, System Verilog, UVM, Testbench Development, Cadence, Synopsys VCS, Primitime, Quartus II IDE, MATLAB, CAD, Proteus, P-spice, Simplify3D, MobaXterm, TCL, Perl, Basic C, G-Code, PCI-E, PCI, Intel DE-10 Lite FPGA Board, Oscilloscopes, DMM, Signal Generators, Sensors, Probes, Bread Board

PROJECTS

64-bit PCI to SRAM Memory Card

Aug 2023 - Dec 2023

- Prepared timing diagrams for both 64-bit and 32-bit bus transactions, ensuring compliance with industry standards and increasing data integrity by 30%.

Vending Machine

Aug 2023 - Dec 2023

- Integrated finite state machines, including idle, selection, payment, dispensing, and reset states, which improved user interaction efficiency by 25%.
- Tested and verified the design using Verilog testbenches and simulation environments, achieving a 98% success rate in hardware validation on an Intel DE-10 Lite FPGA board.

6-Bit Segmented DAC

Jan 2024 - May 2024

- Added two bits to reduce glitch noise, achieving expected results for both theoretical and simulated values of 12V, enhancing output stability by 40%.

STA for Microprocessor Netlist

Aug 2024 - Dec 2024

- Identified timing violations using reports and optimized the generated slack to improve overall design performance, reducing timing issues by 50%.

4-Bit Ripple Carry Adder

Aug 2024 - Dec 2024

- Generated timing reports by giving design constraints with the help of switches such as from, through, to and max paths for static timing analysis, ensuring 100% accuracy in functionality verification.